

YUTIAN KERIYA, China

WMO: 519310

Lat: 36.850N

Lon: 81.650E

Elev: 1554

StdP: 84.00

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-11.2	-20.7	0.7	-3.8	-19.0	0.8	-3.6	6.1	1.7	4.2	-1.6	0.8	N/A	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	34.9	19.2	33.4	19.1	32.0	18.4	22.2	31.4	20.9	30.7	19.7	29.7	1.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	17.1	29.4	17.6	15.3	27.7	16.2	13.9	25.4	74.4	31.0	68.7	30.7	64.0	29.6	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.2	3.4	DB	-16.0	37.6	3.0	1.1	-18.2	38.4	-20.0	39.0	-21.7	39.6	-23.9	40.4	(4)
(5)			WB	-16.1	25.5	2.0	1.0	-17.6	26.2	-18.8	26.8	-19.9	27.3	-21.4	28.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-4.4	0.3	7.5	14.5	19.4	22.6	24.5	23.4	18.6	11.5	4.2	-3.0	(6)	
	DBStd	10.54	3.05	3.01	3.70	3.39	3.27	2.52	2.24	2.58	2.53	3.46	3.55	3.59	(7)	
	HDD10.0	1419	446	271	94	4	1	0	0	0	0	24	176	403	(8)	
	HDD18.3	3017	704	504	335	120	26	2	0	2	27	212	424	661	(9)	
	CDD10.0	2024	0	0	18	141	293	379	450	416	257	70	1	0	(10)	
	CDD18.3	582	0	0	0	6	59	131	192	159	34	0	0	0	(11)	
	CDH23.3	6916	0	0	0	187	753	1451	2154	1796	528	46	0	0	(12)	
	CDH26.7	2797	0	0	0	46	236	569	1011	802	129	4	0	0	(13)	
(14) Wind	WSAvg	1.3	1.2	1.3	1.6	1.7	1.5	1.5	1.2	1.0	1.1	1.1	1.2	1.0	(14)	
(15) Precipitation	PrecAvg	60	2	2	3	3	6	15	11	7	8	1	0	1	(15)	
	PrecMax	184	9	9	14	34	29	85	44	71	69	7	3	4	(16)	
	PrecMin	16	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	38	3	2	4	9	8	18	11	14	18	2	1	1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	14.1	21.8	30.7	33.3	36.0	36.6	36.4	31.7	26.9	19.2	10.8	(19)	
		MCWB	0.4	3.8	8.0	13.8	15.9	18.1	19.4	20.5	16.2	11.7	7.1	2.1	(20)	
	2%	DB	4.7	10.5	19.7	27.6	31.1	33.8	34.9	34.4	29.6	24.5	16.1	6.9	(21)	
		MCWB	-1.4	2.0	7.1	12.2	15.6	17.7	19.5	20.1	16.0	11.3	5.5	0.3	(22)	
	5%	DB	2.9	8.5	17.9	25.5	29.4	31.7	33.6	33.0	28.1	22.0	13.8	4.6	(23)	
		MCWB	-2.3	0.8	6.4	11.1	14.8	16.8	19.4	19.9	15.2	10.3	4.7	-0.9	(24)	
	10%	DB	1.2	6.6	15.7	23.2	27.7	30.2	31.9	31.3	26.5	19.8	11.5	2.6	(25)	
		MCWB	-3.3	-0.1	5.4	9.8	14.2	16.4	18.9	19.2	14.2	9.1	3.7	-2.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	4.2	8.7	14.7	19.1	21.0	23.8	24.7	20.4	13.7	7.6	2.5	(27)	
		MCDB	7.5	12.8	20.2	29.2	29.8	31.1	32.1	32.5	28.0	22.0	17.1	10.5	(28)	
	2%	WB	-1.3	2.2	7.6	12.9	17.1	19.2	22.1	22.7	18.1	12.1	6.1	0.4	(29)	
		MCDB	3.9	9.8	18.8	26.6	28.5	30.5	31.6	31.6	26.6	23.1	15.4	6.1	(30)	
	5%	WB	-2.2	1.0	6.6	11.5	15.9	18.1	20.8	21.3	16.7	10.9	4.9	-0.7	(31)	
		MCDB	2.4	7.7	17.3	24.2	27.1	29.5	31.1	30.7	25.9	21.7	13.4	4.1	(32)	
	10%	WB	-3.1	0.0	5.6	10.1	14.7	17.1	19.6	20.1	15.4	9.4	3.8	-1.8	(33)	
		MCDB	1.1	6.2	15.3	22.0	26.3	28.2	29.9	29.8	24.7	18.5	11.4	2.5	(34)	
(35) Mean Daily Temperature Range	MDBR		10.4	10.7	12.6	13.8	13.4	13.2	13.2	13.9	16.1	16.2	12.9	10.2	(35)	
	5% DB	MCDBR	12.7	14.2	15.7	18.7	17.6	16.8	17.3	18.2	18.8	18.9	16.8	13.5	(36)	
		MCWBR	8.1	7.7	7.4	8.7	9.0	7.5	7.9	9.5	8.5	9.0	8.9	8.2	(37)	
	5% WB	MCDBR	12.2	13.1	14.9	17.1	14.3	14.3	15.0	16.6	16.1	18.9	16.8	12.6	(38)	
		MCWBR	7.9	7.2	7.3	8.3	8.2	7.4	8.0	10.0	8.9	9.1	8.8	7.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.314	0.374	0.484	0.581	0.614	0.632	0.626	0.641	0.576	0.438	0.356	0.326	(40)	
	taud		2.179	1.998	1.718	1.544	1.534	1.551	1.588	1.553	1.634	1.859	2.058	2.130	(41)	
	Ebn at Noon		858	837	769	720	707	694	693	668	681	751	791	809	(42)	
	Edh at Noon		109	149	218	274	282	277	265	269	235	169	122	107	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.92	3.81	5.20	6.12	6.74	6.57	6.21	5.77	5.31	4.72	3.46	2.78	(44)	
	RadStd		0.29	0.24	0.29	0.24	0.33	0.33	0.50	0.47	0.30	0.12	0.17	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (2 neighbors)		+0.57	N/A	N/A	N/A	-0.96	-1.28	N/A	-140	+151	+61	(47)	

Nomenclature: See separate page